

## 20.4 Cook's distance, DFFITS and DFBETAS

An outlier is a case the model fits badly; an **influential** case is one whose removal would change the conclusions. A remote case that follows the trend of the others exactly has no influence, and a large residual near the centroid moves the fit little. Influence measures, all from Theorem 20.3.1, quantify different consequences of deleting a case.

### 20.4.1 Four measures

#### Definition 20.4.1 — Cook's distance, DFFITS, DFBETAS

Let  $h_{ii} < 1$ , and let  $c_{jj}$  be the  $j$ th diagonal entry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , so that  $s^2 c_{jj}$  is the estimated variance of  $\hat{\beta}_j$ .

(a) **Cook's distance** is

$$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta}_{(i)} - \hat{\beta})}{p s^2}.$$

(b) **DFFITS** is the change in case  $i$ 's own fitted value in units of its standard error,

$$\text{DFFITS}_i = \frac{\hat{y}_i - \mathbf{x}_{(i)}^\top \hat{\beta}_{(i)}}{s_{(i)} \sqrt{h_{ii}}}.$$

(c) **DFBETAS** is the change in one coefficient in units of its standard error,

$$\text{DFBETAS}_{ij} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{s_{(i)} \sqrt{c_{jj}}}.$$

(d) **COVRATIO** compares the generalized variances of the estimates with and without case  $i$ ,

$$\text{COVRATIO}_i = \frac{\det(s_{(i)}^2 (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1})}{\det(s^2 (\mathbf{X}^\top \mathbf{X})^{-1})}.$$

Cook (1977) proposed  $D_i$ ; DFFITS, DFBETAS and COVRATIO are from Belsley, Kuh and Welsch (1980). DFFITS and DFBETAS use  $s_{(i)}$ , for the same reason as  $t_i$ . None requires a refit.

#### Theorem 20.4.1 — Influence identities

Let  $h_{ii} < 1$  and  $n - p \geq 2$ , and let  $\tilde{\mathbf{x}}_j$  and  $\ell_{ij}$  be as in Definition 20.2.1.

$$(a) \quad D_i = \frac{\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)}\|^2}{p s^2} = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}.$$

$$(b) \quad \text{DFFITS}_i = t_i \sqrt{h_{ii}/(1 - h_{ii})}, \text{ and } D_i = \text{DFFITS}_i^2 s_{(i)}^2 / (p s^2).$$

(c)  $\text{DFBETAS}_{ij} = t_i \frac{\tilde{x}_{ij}}{\|\tilde{\mathbf{x}}_j\| \sqrt{1 - h_{ii}}}$ , so  $\text{DFBETAS}_{ij}^2 = t_i^2 \ell_{ij} / (1 - h_{ii})$ .

(d) For every  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,

$$\frac{(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{(i)})^2}{s_{(i)}^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}} \leq \text{DFFITS}_i^2,$$

with equality at  $\boldsymbol{\lambda} = \mathbf{x}_{(i)}$ . In particular  $|\text{DFBETAS}_{ij}| \leq |\text{DFFITS}_i|$  for every  $j$ .

(e)  $\text{COVRATIO}_i = \frac{1}{1 - h_{ii}} \left( \frac{n - p}{n - p - 1 + t_i^2} \right)^p$ .

*Proof.* Put  $\mathbf{d} = \hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} g$  with  $g = \hat{\varepsilon}_i / (1 - h_{ii})$  (Theorem 20.3.1(b)).

(a)  $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)} = \mathbf{X} \mathbf{d}$ , and  $\|\mathbf{X} \mathbf{d}\|^2 = \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d} = g^2 \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = g^2 h_{ii}$ . So  $p D_i s^2 = \hat{\varepsilon}_i^2 h_{ii} / (1 - h_{ii})^2$ , and dividing by  $s^2$  and using  $r_i^2 = \hat{\varepsilon}_i^2 / \{s^2(1 - h_{ii})\}$  gives (a).

(b) The numerator of  $\text{DFFITS}_i$  is  $\mathbf{x}_{(i)}^\top \mathbf{d} = h_{ii} g$ , so  $\text{DFFITS}_i = \sqrt{h_{ii}} \hat{\varepsilon}_i / \{s_{(i)}(1 - h_{ii})\} = t_i \sqrt{h_{ii} / (1 - h_{ii})}$ . Then  $\text{DFFITS}_i^2 s_{(i)}^2 = \hat{\varepsilon}_i^2 h_{ii} / (1 - h_{ii})^2 = p D_i s^2$ .

(c) By Theorem 6.6.2(a) the  $j$ th row of  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is  $\tilde{\mathbf{x}}_j^\top / \|\tilde{\mathbf{x}}_j\|^2$ , so the  $j$ th entry of  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$ , the  $i$ th column of that matrix, is  $\tilde{x}_{ij} / \|\tilde{\mathbf{x}}_j\|^2$ . By Eq. 6.6.1,  $c_{jj} = 1 / \|\tilde{\mathbf{x}}_j\|^2$ . Hence

$$\text{DFBETAS}_{ij} = \frac{\tilde{x}_{ij} g \|\tilde{\mathbf{x}}_j\|^{-2}}{s_{(i)} \|\tilde{\mathbf{x}}_j\|^{-1}} = \frac{\tilde{x}_{ij} \hat{\varepsilon}_i}{s_{(i)} \|\tilde{\mathbf{x}}_j\| (1 - h_{ii})},$$

which is the stated form.

(d)  $\boldsymbol{\lambda}^\top \mathbf{d} = g \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$ . Let  $\mathbf{K}^{1/2}$  be the symmetric square root of  $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$  (Theorem 1.7.6). The Cauchy–Schwarz inequality (Proposition 1.1.1) for  $\mathbf{K}^{1/2} \boldsymbol{\lambda}$  and  $\mathbf{K}^{1/2} \mathbf{x}_{(i)}$  gives  $(\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)})^2 \leq \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \cdot h_{ii}$ , with equality when  $\boldsymbol{\lambda}$  is a multiple of  $\mathbf{x}_{(i)}$ . So the ratio is at most  $g^2 h_{ii} / s_{(i)}^2 = \text{DFFITS}_i^2$ . Taking  $\boldsymbol{\lambda} = \mathbf{e}_j$  gives  $\text{DFBETAS}_{ij}^2$ .

(e) By Example (Deleting an observation),  $\det(\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)}) = (1 - h_{ii}) \det(\mathbf{X}^\top \mathbf{X})$ , so  $\text{COVRATIO}_i = (s_{(i)}^2 / s^2)^p / (1 - h_{ii})$ . From the proof of Theorem 20.3.3(c),  $s_{(i)}^2 / s^2 = (n - p - r_i^2) / (n - p - 1)$ , and solving  $t_i^2 = r_i^2 (n - p - 1) / (n - p - r_i^2)$  for  $r_i^2$  gives  $n - p - r_i^2 = (n - p)(n - p - 1) / (n - p - 1 + t_i^2)$ . Substitute.

■

## 20.4.2 What the measures mean

**Cook's distance** has two readings. By (a) it is the squared distance the fitted vector moves when case  $i$  is deleted, in units of  $p s^2$ . By Theorem 12.2.1 with  $\boldsymbol{\Lambda} = \mathbf{I}$ , the  $100(1 - \alpha)\%$  confidence ellipsoid for  $\boldsymbol{\beta}$  is  $\{\mathbf{b} : (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}}) \leq p s^2 F_\alpha(p, n - p)\}$ , so  $\hat{\boldsymbol{\beta}}_{(i)}$  lies on the boundary of the ellipsoid of level  $\Pr\{F(p, n - p) \leq D_i\}$ .

By the second equality in (a),  $D_i$  is misfit,  $r_i^2$ , times leverage,  $h_{ii} / (1 - h_{ii})$ : a case is influential only if it has both. DFFITS factors the same way with  $t_i$ . The **influence plot** of Figure

20.4.1 places each case by leverage and studentized residual, with the curves of constant  $D_i$ ,  $r^2 = pD(1 - h)/h$ .

**DFBETAS** isolates one coefficient: by (c) the partial leverage  $\ell_{ij}$  replaces the leverage in the product, but the factor  $1/(1 - h_{ii})$  still involves the overall leverage. By (d), the largest standardized change in *any* linear combination  $\lambda^\top \hat{\beta}$  is  $|\text{DFFITS}_i|$ , so a small DFFITS rules out a large change in every coefficient and contrast. **COVRATIO** measures precision rather than location: by (e) it exceeds one when  $h_{ii}$  is large and  $t_i$  small (the case adds precision) and falls below one when  $t_i$  is large (the case inflates  $s^2$ ).

### 20.4.3 The null law of Cook's distance and the choice of cut-offs

How large must an influence measure be before it matters? Since  $D_i$  is a multiple of  $r_i^2$ , its law under the model is known exactly.

#### Proposition 20.4.2 — Cook's distance under the model

Assume Eq. 7.3.1 with  $n - p \geq 2$  and all leverages below one. Then  $D_i = \frac{n - p}{p} \frac{h_{ii}}{1 - h_{ii}} B_i$  with  $B_i \sim \text{Beta}(\frac{1}{2}, \frac{n - p - 1}{2})$ , so

$$E(D_i) = \frac{h_{ii}}{p(1 - h_{ii})}, \quad \frac{1}{n} \sum_{i=1}^n E(D_i) \geq \frac{1}{n - p},$$

with equality in the last bound iff all leverages equal  $p/n$ .

*Proof.* Combine Theorem 20.4.1(a) with Proposition 20.1.1(a), and use  $E(B_i) = 1/(n - p)$ . For the bound,  $u \mapsto u/(1 - u)$  is strictly convex on  $[0, 1)$ , so by Jensen's inequality the average of  $h_{ii}/(1 - h_{ii})$  is at least  $\bar{h}/(1 - \bar{h}) = p/(n - p)$ , with equality iff the  $h_{ii}$  are equal. ■

So under the model a typical  $D_i$  is of order  $1/n$ , which explains the popular cut-off  $4/n$ . For the state design of Example (How much does the District of Columbia matter?) the average of  $E(D_i)$  is 0.0240, against the bound  $1/(n - p) = 0.0213$ . Comparing  $D_i$  with the median of  $F(p, n - p)$ , which is near one, is far stricter: it flags only cases whose deletion moves the estimate to the edge of a 50% confidence region. The usual cut-offs are these.

| Measure                 | Cut-off                                | Reasoning                                                     |
|-------------------------|----------------------------------------|---------------------------------------------------------------|
| $h_{ii}$                | $2p/n$                                 | twice the average leverage                                    |
| $D_i$                   | $4/n$ , or the median of $F(p, n - p)$ | four times the typical value; or the edge of the 50% region   |
| $ \text{DFFITS}_i $     | $2\sqrt{p/n}$                          | $ t_i  \approx 2$ at a case of average leverage $p/n$         |
| $ \text{DFBETAS}_{ij} $ | $2/\sqrt{n}$                           | $ t_i  \approx 2$ at a case of average partial leverage $1/n$ |

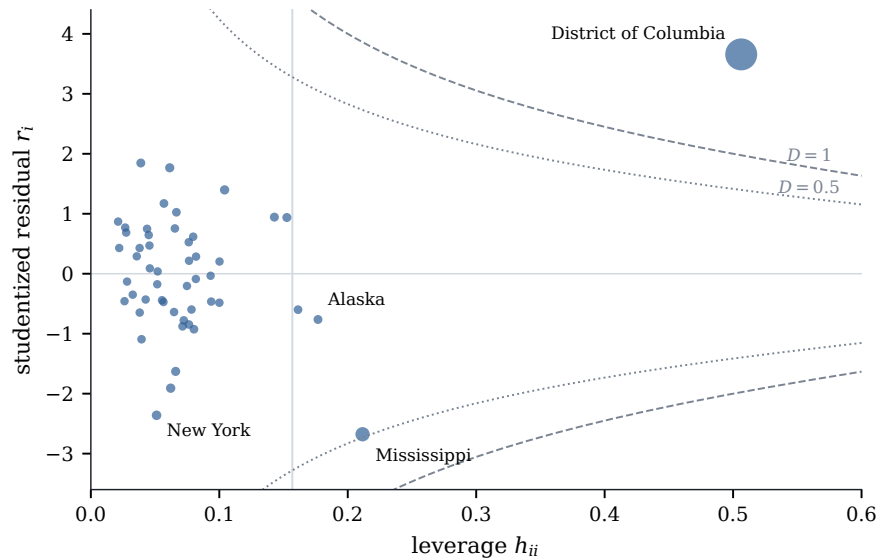


Figure 20.4.1: Influence plot for the state regression: leverage against  $r_i$ , area proportional to Cook's distance, curves  $D_i = 0.5$  (dotted) and  $D_i = 1$  (dashed), and the line  $h_{ii} = 2p/n$ .

| Measure                   | Cut-off | Reasoning                                                                  |
|---------------------------|---------|----------------------------------------------------------------------------|
| $ \text{COVRATIO}_i - 1 $ | $3p/n$  | first-order expansion of (e) at $ t_i  \approx 2$ or $h_{ii} \approx 2p/n$ |

All are arbitrary in the same way: each puts a case with studentized residual about two and typical leverage near the line. Calibrating  $D_i$  exactly by [Proposition 20.4.2](#) would not help: for a fixed design  $D_i$  is a multiple of  $r_i^2$ , so the resulting test is an outlier test with a case-dependent level, discarding the leverage factor that is the point of an influence measure. Influence is a consequence to be assessed, not a hypothesis: the right threshold is whether deleting the case changes a conclusion.

#### Example. (How much does the District of Columbia matter?)

In the murder-rate regression with all 51 jurisdictions, the District of Columbia has  $D_i = 3.430$ , seven times the next largest (Mississippi, 0.481); the median of  $F(4, 47)$  is 0.851, and deleting the District moves  $\hat{\beta}$  to the boundary of the 0.985 confidence ellipsoid. Its DFFITS is 4.333 (cut-off 0.560) and its DFBETAS for poverty  $-1.090$  (cut-off 0.280). Its partial leverage for poverty is only 0.032 (Example (Where the District of Columbia's leverage comes from)), but by [Theorem 20.4.1\(c\)](#) it is multiplied by  $t_i^2/(1 - h_{ii})$ :  $4.278^2 \times 0.032/(1 - 0.506) \approx 1.19$ , so  $|\text{DFBETAS}| \approx 1.09$ . Alaska, with the largest partial leverage (0.141), has  $t_i = -0.760$  and DFBETAS 0.315.

The consequences are substantive. With the District, the poverty coefficient is 0.1551 (standard error 0.1059,  $t = 1.47$ ); without it, 0.2538 (standard error 0.0934,  $t = 2.72$ ), the fit of Example (Poverty and murder rates). One jurisdiction decides whether poverty has a detectable association with murder rates given the other regressors, and the single-parent coefficient moves from 0.6256 to 0.3980. The cut-offs flag 2 cases by  $D_i > 4/n = 0.078$ , 3 by DFFITS, 5 by DFBETAS for poverty and 4 by leverage above  $2p/n = 0.157$ , but only the District changes a conclusion. Its COVRATIO is 0.578: it inflates the estimated generalized variance through its contribution to  $s^2$ .

```

import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statcrime.load_pandas().data           # District of Columbia
    kept
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
names = np.array(data.index)
n, p = X.shape
Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta                                         # raw residuals
h = np.sum(Q ** 2, axis=1)                               # leverages
s2 = e @ e / (n - p)
r = e / np.sqrt(s2 * (1 - h))                           # internally studentized
s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1)      # s^2 without case i
t = e / np.sqrt(s2_del * (1 - h))                       # externally studentized

cook = r ** 2 * h / (p * (1 - h))                       # Cook's distance
dffits = t * np.sqrt(h / (1 - h))
K = np.linalg.inv(X.T @ X)
C = K @ X.T                                              # row j: weights of y in beta_j
dfbetas = (C * e / (1 - h)).T / np.sqrt(np.outer(s2_del, np.diag(K)))
for i in np.argsort(-cook)[:3]:
    print(f"{names[i]:22s} D = {cook[i]:.3f}  DFFITS = {dffits[i]:6.3f}"
          f"  DFBETAS(poverty) = {dfbetas[i, 1]:6.3f}")
print("conventional cut-offs: D > 4/n =", round(4 / n, 3),
      " |DFFITS| > 2 sqrt(p/n) =", round(2 * np.sqrt(p / n), 3),
      " |DFBETAS| > 2/sqrt(n) =", round(2 / np.sqrt(n), 3))

```

[Run this code online](#)

## 20.4.4 Exercises

### 20.4.4.1 A. Check your understanding

#### Exercise A1.

In a regression with  $p = 3$ , case  $i$  has  $r_i = 2$ . Compute  $D_i$  when  $h_{ii} = 0.1$  and when  $h_{ii} = 0.6$ . Which case would you look at first, and why might neither be alarming?

*Solution.* By [Theorem 20.4.1\(a\)](#),  $D_i = (4/3) h_{ii} / (1 - h_{ii})$ :  $(4/3)(1/9) = 4/27 \approx 0.15$  and  $(4/3)(1.5) = 2$ . The second moves the coefficient vector to about the edge of a high-level confidence region. Neither is alarming by itself if the change it produces in the estimates of interest is small relative to their uncertainty, or if the case is known to be correct and the model to hold in its region.

#### Exercise A2.

Show that  $D_i = 0$  iff  $\hat{\varepsilon}_i = 0$ , whatever the leverage. Give an example of a case with leverage 0.9 and no influence.

## 20.4.4.2 B. Practice

**Exercise B1.**

Let  $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . Show that the sum of squares of the  $j$ th row of  $\mathbf{C}$  is  $c_{jj}$ , and deduce that  $\sum_i \ell_{ij} = 1$  in a second way. Why does this suggest a cut-off of order  $1/\sqrt{n}$  for DFBETAS?

*Solution.*  $\mathbf{C}\mathbf{C}^\top = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1}$ , whose  $j$ th diagonal entry is  $c_{jj}$ . The  $j$ th row of  $\mathbf{C}$  is  $\tilde{\mathbf{x}}_j^\top / \|\tilde{\mathbf{x}}_j\|^2$  (proof of [Theorem 20.4.1\(c\)](#)), so  $\sum_i \ell_{ij} = \sum_i \tilde{x}_{ij}^2 / \|\tilde{\mathbf{x}}_j\|^2 = c_{jj} \|\tilde{\mathbf{x}}_j\|^2 = 1$ . The partial leverages for a coefficient average  $1/n$ . With  $|t_i| \approx 2$  and  $h_{ii}$  small,  $|\text{DFBETAS}_{ij}| \approx 2\sqrt{\ell_{ij}} \approx 2/\sqrt{n}$  for a typical case.

**Exercise B2.**

For simple regression, show that the DFBETAS of case  $i$  for the slope is  $t_i(x_i - \bar{x}) / \sqrt{S_{xx}(1 - h_{ii})}$ . Which cases can change the slope most?

**Exercise B3.**

Use [Theorem 20.4.1\(e\)](#) to show that for large  $n$  and small  $p/n$ ,  $\text{COVRATIO}_i \approx 1 + h_{ii} - p(t_i^2 - 1)/n$ . Deduce the cut-off  $|\text{COVRATIO}_i - 1| > 3p/n$  from the two extreme cases  $h_{ii} \approx 0$ ,  $|t_i| = 2$  and  $h_{ii} = 2p/n$ ,  $t_i = 0$ .

*Solution.*  $1/(1 - h) \approx 1 + h$ , and  $((n - p)/(n - p - 1 + t^2))^p = (1 + (t^2 - 1)/(n - p))^{-p} \approx 1 - p(t^2 - 1)/n$ . Multiplying,  $\text{COVRATIO} \approx 1 + h - p(t^2 - 1)/n$  to first order. With  $h \approx 0$  and  $t^2 = 4$  this is  $1 - 3p/n$ : a case of no leverage and a residual of two standard errors lowers the precision by that much. With  $t = 0$  and  $h = 2p/n$  it is  $1 + 3p/n$ : a well-fitting case of high leverage raises it by the same amount. Deviations beyond  $3p/n$  in either direction are more extreme than both.